

MILESTONE 3

Comprehensive Performance & Deliverables Report

AI-Powered Wildlife Population Intelligence System

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Program	Infosys Springboard Virtual Internship Program
Batch	Batch 2
Project	AI Wildlife Population Intelligence System
Repository	github.com/Jayasrip08/wildlife-population-intelligence-system
Milestone	Milestone 3 (Weeks 5 & 6) — Completed

Executive Summary

Milestone 3 (Weeks 5 & 6) of the AI-Powered Wildlife Population Intelligence System has been fully implemented, integrated, and verified against the official project PDF specification. This milestone adds advanced population abundance estimation, spatial density analytics, satellite vegetation (NDVI) habitat intelligence, and the 5-factor weighted Ecosystem Health Scoring model.

This document consolidates the modules built, the estimation and scoring formulas used, the API surface exposed, the database schema extensions, and the verification results completed during Milestone 3 into a single reference file, following the same reporting structure established in Milestones 1 and 2.

1. Project Overview & SDLC Progress

Continuing the Iterative & Agile SDLC model, Milestone 3 moved the system from per-detection intelligence (Milestone 2) into population-level and landscape-level intelligence — abundance estimation, migration corridor mapping, satellite-derived habitat quality, and conservation decision support.

Milestone Roadmap

- **Milestone 1 (Completed):** Project Initialization, Requirement Analysis, Architecture & Database Design, JWT Authentication & RBAC, Dataset Ingestion Pipeline Setup.
- **Milestone 2 (Completed):** Species Recognition Engine (Vision + Audio), Biodiversity Analytics Engine, PDF Audit Report Generation, Frontend & Backend Integration.
- **Milestone 3 (Completed):** Population Abundance & Migration Engine, Habitat Intelligence (NDVI/Canopy) Engine, Weighted Conservation Health Scoring, Frontend Intelligence Dashboards.

2. System Architecture — Population & Conservation Layer

Milestone 3 extends the Milestone 1/2 architecture (React + Vite frontend, FastAPI backend, PostgreSQL/PostGIS + MongoDB data layer, PyTorch/Librosa detection layer) with a population statistics, remote-sensing habitat analysis, and conservation decision-support layer.

Layer	Component / Stack
Frontend Interface	React + Vite — Population Intelligence, Habitat Intelligence, and Conservation Engine tabs
API Gateway	REST API (JSON / JWT) — FastAPI routers for population, habitat, and conservation
Population Engine	Abundance & migration statistics — backend/ml_population.py
Habitat Engine	NDVI, canopy cover & degradation analysis — backend/habitat.py
Conservation Engine	5-factor weighted health scoring & recommendations — backend/conservation.py
Persistence	PostgreSQL (ORM tables) — population_estimates, habitat_assessments, conservation_recommendations
Data Schemas	SQLAlchemy models & Pydantic schemas — backend/models.py & backend/schemas.py

Component Breakdown

- Population Estimation & Migration Engine: derives total estimated population from observed count and detection probability, computes density per km² with 95% confidence intervals, and maps seasonal migration routes, velocities, bearings, and threat intersections.
- Habitat Intelligence & Vegetation Engine: evaluates satellite NDVI and canopy cover, computes a habitat fragmentation/degradation index, and identifies threat drivers such as deforestation fringes and drought stress.
- Conservation Recommendation & Health Scoring Engine: rolls species diversity, population stability, habitat quality, endangered-species status, and environmental conditions into a single weighted Ecosystem Health Score with a categorical status and prioritized action plan.
- Database Schema Extensions: SQLAlchemy models and Pydantic schemas for population census runs, habitat assessments, and conservation recommendations.
- Interactive Frontend Dashboards: Population Intelligence tab for sample size / survey area inputs and corridor telemetry, Habitat Intelligence tab for NDVI vigor and canopy suitability, and a Conservation Engine tab with interactive 5-factor weighted sliders computing health in real time.

3. Population Intelligence & Conservation Workflow

Milestone 3 implements the population and conservation decision-support stages of the end-to-end wildlife monitoring workflow introduced in Milestone 1:

Stage	Description
1. Survey Input	Sample size, survey area, and detection probability submitted for a monitoring site or corridor.
2. Abundance & Density Estimation	$N = n / p$ computed; density per km ² and 95% confidence interval derived.
3. Migration & Corridor Mapping	Seasonal routes, movement velocity, bearing, and threat intersections plotted per species.
4. Habitat Intelligence	NDVI, canopy cover, fragmentation, and habitat suitability evaluated from satellite/land-cover data.
5. Conservation Scoring & Action Planning	5-factor weighted Ecosystem Health Score computed; status tier and prioritized conservation actions generated.

4. Requirements Matrix & Specifications

A. Functional Requirements (FR)

ID	Category	Description	Status
FR-11	Population Abundance Estimation	Estimate total population ($N = n / p$) and density per km ² via POST <code>/api/v1/population/estimate</code> .	☒ Completed
FR-12	Migration Corridor Telemetry	Serve migration routes, velocity, bearing, and threat intersections via GET <code>/api/v1/population/migration/{species}</code> .	☒ Completed
FR-13	Habitat & Vegetation Analysis	Assess NDVI, canopy cover, degradation index, and suitability via POST <code>/api/v1/habitat/analyze</code> .	☒ Completed
FR-14	Conservation Health Scoring	Compute the 5-factor weighted Ecosystem Health Score and status tier via POST <code>/api/v1/conservation/recommendations</code> .	☒ Completed
FR-15	Frontend Intelligence Dashboards	Population Intelligence, Habitat Intelligence, and Conservation Engine tabs with real-time interactivity.	☒ Completed

B. Non-Functional Requirements (NFR)

ID	Attribute	Target Metric / Specification	Verification
NFR-11	Statistical Accuracy	Abundance and density formulas follow the PDF specification exactly ($N = n / p$; 95% CI).	Verified
NFR-12	Spec Conformance	Ecosystem Health Score weighting matches the official 30/25/20/15/10 PDF formula.	Verified
NFR-13	Data Integrity	Population, habitat, and conservation results persist to normalized PostgreSQL ORM tables.	Verified
NFR-14	Build Reliability	Frontend production build compiles cleanly with zero errors.	Pass
NFR-15	Usability	Real-time slider-driven health scoring and corridor telemetry without manual refresh.	Verified

5. User Stories & Acceptance Criteria

User Story 1: Wildlife Researcher — Population Abundance Estimation

- As a Wildlife Researcher
- I want to enter observed counts and survey area to receive an estimated population size, density, and confidence interval
- So that I can report defensible population figures without manual statistical calculation.

Acceptance Criteria

- Population Intelligence tab accepts sample size and survey area, and returns N , density per km^2 , and a 95% confidence interval.
- Migration corridor view shows movement velocity, bearing, and any threat intersections for the selected species.

User Story 2: Habitat Analyst — Vegetation & Degradation Assessment

- As a Habitat Analyst
- I want to run satellite-based NDVI and canopy cover analysis on a monitoring zone
- So that I can flag habitat degradation and fragmentation before it threatens resident species.

Acceptance Criteria

- Habitat Intelligence tab displays NDVI value, canopy cover percentage, and a habitat suitability score.
- Degradation index highlights key threat drivers such as deforestation fringes or drought stress.

User Story 3: Conservation Manager — Weighted Health Scoring & Action Planning

- As a Conservation Manager
- I want a single weighted ecosystem health score with a clear status tier and prioritized actions

- So that I can allocate anti-poaching patrols, sensor placements, and restoration efforts where they matter most.

Acceptance Criteria

- Conservation Engine tab computes health score in real time from five weighted factors and classifies it as Excellent, Healthy, Moderate Concern, Vulnerable, or Critical.
- Recommendations list concrete priority actions, e.g. patrol unit allocation, water-source sensor placement, or restoration corridors.

6. Database Schema Extensions (SQLAlchemy / PostgreSQL)

Milestone 3 introduces three new relational models to persist population census runs, habitat assessments, and conservation recommendations, extending the schema defined in Milestones 1 and 2.

1. Table: population_estimates

```
CREATE TABLE population_estimates (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  survey_id UUID REFERENCES surveys(id) ON DELETE CASCADE,
  species VARCHAR(150),
  observed_count INTEGER NOT NULL,          -- n
  detection_probability NUMERIC(4,3) DEFAULT 0.82, -- p
  estimated_population INTEGER,              -- N = n / p
  survey_area_km2 NUMERIC(10,2),
  density_per_km2 NUMERIC(8,3),
  ci_lower_bound NUMERIC(10,2),              -- 95% CI
  ci_upper_bound NUMERIC(10,2),
  growth_trend VARCHAR(20),                  -- Increasing | Stable | Declining
  computed_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

2. Table: habitat_assessments

```
CREATE TABLE habitat_assessments (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  site_id UUID REFERENCES monitoring_sites(id) ON DELETE CASCADE,
  ndvi_value NUMERIC(4,3),
  canopy_cover_percent NUMERIC(5,2),
  degradation_index NUMERIC(5,2),
  threat_driver VARCHAR(100),                 -- e.g. Deforestation Fringe, Drought Stress
  land_cover_class VARCHAR(50),
  suitability_score NUMERIC(5,2),
  assessed_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

3. Table: conservation_recommendations

```
CREATE TABLE conservation_recommendations (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  survey_id UUID REFERENCES surveys(id) ON DELETE CASCADE,
  species_diversity_score NUMERIC(5,2),    -- 30%
  population_stability_score NUMERIC(5,2),  -- 25%
  habitat_quality_score NUMERIC(5,2),       -- 20%
  endangered_status_score NUMERIC(5,2),     -- 15%
  environmental_conditions_score NUMERIC(5,2),-- 10%
  ecosystem_health_score NUMERIC(5,2),
  status_tier VARCHAR(20) CHECK (status_tier IN
    ('Excellent','Healthy','Moderate Concern','Vulnerable','Critical')),
  urgency_level VARCHAR(20),
  priority_actions TEXT,
  generated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

7. Milestone 3 Benchmarks & Verification Summary

PDF Requirements vs. Implementation

PDF Requirement	Status	File Path
Population Counting & Density	 Completed	backend/ml_population.py
Migration Corridor Telemetry	 Completed	backend/ml_population.py
Vegetation (NDVI) & Canopy Analysis	 Completed	backend/habitat.py
Degradation Detection	 Completed	backend/habitat.py
Weighted Ecosystem Health Score	 Completed	backend/conservation.py
Conservation Priority Actions	 Completed	backend/conservation.py

Backend API & Build Verification

Check	Description	Result
GET /health	System telemetry check (FastAPI v3.0.0)	{"status": "ok", "milestone": 3}
POST /api/v1/population/estimate	Abundance & density estimation	Verified
GET /api/v1/population/migration/{species}	Migration corridor telemetry	Verified
POST /api/v1/habitat/analyze	NDVI / canopy / degradation analysis	Verified
POST /api/v1/conservation/recommendations	Weighted health score & recommendations	Verified
npx vite build	Frontend production build	100% clean, zero errors (dist/assets/index-DBw1moIC.js)

Backend verified via FastAPI health_check() reporting version 3.0.0 and milestone 3; frontend production build completed with a clean compile and no errors.

8. Environment Setup Reference (.env) — Milestone 3 Additions

The following environment variables extend the Milestone 1/2 configuration template with settings required for population statistics and satellite habitat analysis.

```
# Population Estimation Settings
DEFAULT_DETECTION_PROBABILITY=0.82
CONFIDENCE_INTERVAL_LEVEL=0.95

# Habitat / Satellite Data
NDVI_DATA_SOURCE=sentinel-2
NDVI_CACHE_PATH=./cache/ndvi
CANOPY_MODEL_PATH=./models/canopy_cover_model.pt

# Conservation Scoring Weights
WEIGHT_SPECIES_DIVERSITY=0.30
WEIGHT_POPULATION_STABILITY=0.25
WEIGHT_HABITAT_QUALITY=0.20
WEIGHT_ENDANGERED_STATUS=0.15
WEIGHT_ENVIRONMENTAL_CONDITIONS=0.10
```

Note: satellite imagery credentials and cached NDVI tiles are git-ignored in the submission copy; only path and weighting configuration is shown here.

Conclusion & Project Outlook

Milestone 3 successfully delivered the population and conservation decision-support layer of the Wildlife Population Intelligence System — abundance and density estimation, migration corridor mapping, satellite-based habitat intelligence, and the 5-factor weighted Ecosystem Health Score with prioritized conservation actions — fully integrated with the Milestone 1 and Milestone 2 foundations and verified end-to-end against the official PDF specification.

With population, habitat, and conservation intelligence now complete alongside the earlier detection and biodiversity engines, the system delivers a full pipeline from raw field data to actionable conservation guidance, positioning the project for production hardening, field pilot deployment, and any further planned enhancements.